

ProvBench-RAG: Provenance-Sensitive Retrieval Under Near-Duplicate Source Confusion

Anonymous CIKM submission

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Abstract

Retrieval-augmented generation (RAG) systems are often evaluated on whether an answer can be supported by *some* retrieved passage. Deployed assistants, however, must routinely choose among near-duplicate sources—mirrors, stale snapshots, scraped copies, and jurisdiction-specific variants—that share topical overlap but differ in authority, validity, and attribution. We study this *instance-level provenance* problem under lexical collision: many candidates look equally relevant, yet only one (or none) should be cited.

We present **ProvBench-RAG**, a benchmark specification and reproducible dataset format with clustered documents, gold preferred sources, minimal evidence sets, temporal and jurisdiction constraints, and abstention when authoritative sources disagree. We further introduce **PACER** (*Provenance-Aware Cluster Evidence Reranking*), a transparent reranker that fuses min-max normalized BM25 scores with declarative provenance signals (authority, freshness, originality/copy labels, jurisdiction alignment) over a fixed candidate pool.

On a controlled synthetic split of 1500 queries with explicit lexical decoys, BM25 and semantic-only reranking achieve $\text{PSA}@1 = 0$ while PACER reaches 0.750 (McNemar $p < 10^{-200}$, $n=1200$), with ProvScore 0.469→0.703 and nDCG@k 0.553→0.926. Stratified analysis shows gains across freshness, authority, and jurisdiction families, not only copy detection. On 140 English Wikipedia revision clusters, PACER improves $\text{PSA}@1$ from 0.014 to 0.057 ($p=0.03125$), though absolute provenance accuracy remains low, highlighting the difficulty of real revision corpora. A pooled heterogeneous index (1640 queries) demonstrates recall collapse under a single lexical index (0.435) and partial recovery with provenance reranking (0.782). All tables and macros regenerate from the released artifact.

1 Introduction

Large language models augmented with retrieval [2, 6] are deployed over corpora where redundancy is structural rather than exceptional. Regulatory portals, vendor documentation, and encyclopedic content routinely contain multiple passages that answer the same surface query with incompatible provenance: an SEO summary may repeat the user’s phrasing while citing the wrong deadline; a mirrored manual may read as current while its validity window has expired; two agencies may publish equally authoritative yet contradictory guidance.

In these settings, “the model found supporting text” is insufficient. Downstream generation and citation interfaces require selecting the *right document instance*—the correct revision, issuer, and jurisdiction—before faithfulness to an extracted span is even meaningful. We call the resulting retrieval failure mode **near-duplicate source confusion**: candidates rank together under lexical or dense similarity, but provenance metadata determines which instance is safe to attribute.

Existing RAG evaluation emphasizes answer correctness, grounding to supplied context, multi-hop retrieval, or version graphs over evolving documents [3–5]. These efforts only partially cover

Table 1: Positioning against recent benchmarks (✓: explicit task coverage).

Work	Answer grounding	Preferred <i>instance</i>	Time / version axis	Explicit conflict
FACTS [4]	✓	–	–	–
FRAMES [5]	✓	implicit (multi-source hops)	partial	limited
VersionRAG [3]	✓	version graph	✓	–
ProvBench-RAG	✓	✓ (P^*)	✓	✓

the decision we target: *which sibling instance should rank first* when near-duplicates disagree on authority, time, originality, or jurisdiction (Table 1). ProvBench-RAG makes that decision measurable with explicit cluster structure and instance-level labels, rather than treating retrieved context as an undifferentiated bundle.

We also describe PACER, an intentionally simple and auditable reranker that operates on a BM25 candidate pool and combines lexical relevance with normalized provenance features. PACER is not proposed as a production ranker; it is a diagnostic baseline that exposes when declarative metadata is sufficient to break lexical ties, and where additional modeling (e.g., cross-encoders or learned fusion) remains necessary.

Contributions.

1. **ProvBench-RAG.** A JSONL benchmark specification with clustered near-duplicates, preferred-source annotations, minimal evidence, and conflict abstention, plus released synthetic, Wikipedia revision, and pooled tracks.
2. **PACER.** A transparent provenance-aware reranker with ablatable signal terms and cluster-level variants, designed for controlled comparison against lexical and heuristic baselines.
3. **Evaluation protocol.** A composite **ProvScore** metric, sanity controls (random/oracle pool bounds), stratified analysis by confusion family, and paired significance testing with artifact-driven table generation.

2 Related work

Retrieval-augmented generation. RAG couples parametric memory with non-parametric retrieval [6]; surveys document retrieval, grounding, and temporal drift as first-class concerns [2]. Most benchmarks nonetheless score whether an answer is supported by retrieved content in aggregate, without requiring the retriever to prefer a specific mirror or revision when several passages are mutually redundant.

Grounding and unified RAG evaluation. FACTS [4] measures faithfulness to long-form input but does not annotate competing near-duplicate instances. FRAMES [5] unifies factuality, retrieval, and reasoning across multi-hop settings, yet does not specify which source should win under lexical collision across mirrors.

Version-aware retrieval. VersionRAG [3] models evolving documents with explicit version graphs, aligning with our freshness axis but not with jurisdiction mismatch, copy farms, or authority-copy interactions on static snapshots.

Fusion and reranking. Classical fusion methods such as reciprocal rank fusion (RRF) [1] combine ranked lists without feature transparency. We include $\text{RRF}_{\text{BM25+Auth}}$ as a non-learned orchestration baseline alongside single-signal heuristics.

ProvBench-RAG complements prior work by isolating instance-level provenance decisions under near-duplicate lexical overlap, rather than end-to-end answer correctness alone.

3 Task, metrics, and annotations

Each query q specifies a gold answer y^* , gold evidence set E^* , minimal evidence set $M^* \subseteq E^*$, preferred sources P^* (when defined), query time and jurisdiction constraints, and an abstention flag for irreconcilable authoritative conflict. Documents belong to clusters of near-duplicates; cluster metadata records relation type (e.g., original vs. copy, fresh vs. stale).

The system outputs a ranked list $R(q)$ of document identifiers. Synthetic queries include explicit **Answer**: spans in document text; the Wikipedia track aligns gold answers with the first sentence of the newest revision under programmatic validity windows [7].

Retrieval and provenance metrics. We report $\text{recall}@k$, $\text{nDCG}@k$, **preferred-source accuracy at rank 1** (PSA@1) on non-abstaining queries, and provenance F1@3 over the top three ranks when P^* is defined. Answer correctness and abstention behavior are evaluated with deterministic span extractors over the top-ranked passage, enabling full reproducibility.

ProvScore. End-to-end provenance quality is summarized as:

$$\text{ProvScore} = 0.25 \text{ Ans} + 0.25 \text{ PSA} + 0.20 \text{ Suff} + 0.15 \text{ Min} + 0.15 \text{ Abst}, \quad (1)$$

where Ans is answer accuracy, PSA is preferred-source accuracy (neutral 0.5 when undefined in micro-averages), Suff indicates whether E^* appears in the top- k list, Min penalizes retrieving strict supersets of M^* , and Abst scores conflict abstention.

4 Datasets

4.1 Synthetic documentation and policy track

The synthetic generator produces five families of 300 clusters each (1500 queries total): **A** original vs. blog vs. SEO copy with divergent answers; **B** stale archive vs. current manual vs. lexical decoy mirror that echoes query-time phrasing; **C** content farm vs. agency rulebook vs. tertiary summary; **D** US vs. EU regulation under explicit US jurisdiction, including a decoy that repeats jurisdiction keywords incorrectly; **E** two authoritative sources with conflicting answers and abstain gold. Families B–D include *lexical decoys*: high-BM25 passages that mimic query wording but violate freshness, authority, or jurisdiction constraints.

4.2 Wikipedia revision freshness track

Using the public MediaWiki API [7], we sample three revisions per article (oldest, median, newest in a fetched window), construct validity intervals so that only the newest revision applies at the query date, and define P^* as the newest revision document. This track comprises 140 article clusters after filtering missing pages and insufficient revision history.

4.3 Pooled heterogeneous corpus

We concatenate synthetic and Wikipedia JSONL into a single BM25 index without domain routing, simulating a heterogeneous enterprise corpus where query conventions and metadata schemas differ across sources.

5 Methods

All methods share the same inverted index and, for pool-based rankers, the same `top-pool_k` BM25 candidates (`pool_k=80`, $k=10$).

5.1 Sanity controls

`control_random_pool` permutes pool documents with a query-seeded RNG (stochastic lower bound). `control_semantic_only` runs PACER with provenance weights set to zero, isolating lexical ordering within the pool. `control_oracle_pool` promotes the first P^* member in the pool to rank 1, then follows BM25 order (upper bound given pool membership).

5.2 Baselines

BM25 ranks the full corpus by Okapi BM25. **authority_first** and **recent_first** rerank an enlarged BM25 pool by authority level and publication date. **RRF_{BM25+Auth}** fuses BM25 and authority orderings via RRF [1] with $k=60$.

5.3 PACER

For each document d in the BM25 pool, PACER computes min-max normalized semantic score $s_{\text{sem}}(d | q)$ and provenance components s_{auth} , s_{fresh} , s_{orig} , s_{jur} in $[0, 1]$, then

$$\text{PACER}(d | q) = \frac{\alpha s_{\text{sem}} + \beta s_{\text{auth}} + \gamma s_{\text{fresh}} + \delta s_{\text{orig}} + \eta s_{\text{jur}}}{\alpha + \beta + \gamma + \delta + \eta}. \quad (2)$$

Default weights $\alpha=\beta=\gamma=\delta=\eta=1$. **cluster-PACER** ranks clusters by the maximum PACER score within each cluster, emits the best document per cluster, then fills remaining slots from document-level PACER.

5.4 Ablations

We zero one provenance coefficient at a time (`drop_authority`, `drop_freshness`, `drop_originality`, `drop_jurisdiction`) while holding the pool fixed.

6 Experimental protocol

Experiments are fully scripted: `scripts/reproduce.sh` rebuilds synthetic data; `scripts/reproduce_full.sh` adds Wikipedia, merging, combined evaluation, and extended analysis. Significance testing uses exact McNemar tests on paired PSA@1 outcomes (BM25 vs. PACER), bootstrap confidence intervals (2000 resamples) on micro-averaged ProvScore, and Wilcoxon signed-rank tests on paired per-query gains.

Table 2: Synthetic ProvBench-RAG extended comparison ($n=1500$). Prov.F1: macro provenance F1@3 on queries with P^* .

Method	PSA@1	R@ k	nDCG	Prov.F1	Ans	ProvScore
control_random_pool	0.006	0.116	0.060	0.013	0.394	0.275
control_semantic_only	0.000	1.000	0.553	0.375	0.413	0.469
bm25	0.000	1.000	0.553	0.375	0.413	0.469
authority_first	0.750	1.000	0.926	0.500	0.600	0.703
recent_first	0.000	0.200	0.063	0.000	0.404	0.294
rrf_bm25_authority	0.622	1.000	0.877	0.499	0.559	0.661
pacerc	0.750	1.000	0.926	0.500	0.600	0.703
cluster_pacer	0.750	0.900	0.926	0.500	0.600	0.680
control_oracle_pool	1.000	1.000	0.914	0.500	0.589	0.763

Table 3: ProvScore stratification: BM25 vs. PACER by synthetic family.

Stratum	n	BM25	PACER	Δ
A	300	0.448	0.490	+0.042
B	300	0.415	0.698	+0.283
C	300	0.448	0.782	+0.333
D	300	0.415	0.740	+0.325
E	300	0.617	0.617	+0.000

7 Results

7.1 Synthetic primary comparison

Table 2 reports the extended synthetic benchmark. With lexical decoys in families B–D, BM25 and **control_semantic_only** achieve $\text{PSA@1} = 0$ despite perfect recall@k : the top-ranked passage is frequently a decoy that matches query wording. PACER and **authority_first** both reach $\text{PSA@1} = 0.75$ and $\text{ProvScore} = 0.703$, indicating that authority and multi-signal fusion suffice to break many ties once lexical scores are pool-normalized. $\text{RRF}_{\text{BM25+Auth}}$ is competitive ($\text{PSA@1} = 0.622$, $\text{ProvScore} = 0.661$) but does not match full PACER on nDCG and PSA. The oracle pool control reaches $\text{PSA@1} = 1.0$, quantifying headroom when P^* is known to lie in the pool.

7.2 Stratified analysis

Table 3 shows ProvScore by synthetic family. Gains are largest on freshness (B), authority (C), and jurisdiction (D) strata; family A (original vs. copy) improves modestly; abstention-only family E is unchanged by design. This pattern matches the generator’s decoy structure: lexical overlap alone is insufficient, and provenance signals recover the correct instance when decoys optimize surface similarity.

7.3 Ablations

Table 4 shows that removing authority or originality terms reduces ProvScore from 0.703 to 0.634, while dropping freshness or jurisdiction leaves performance unchanged on this generator. Freshness

Table 4: Single-term PACER ablations (synthetic).

Variant	ProvScore
Full PACER	0.703
drop_authority	0.634
drop_freshness	0.703
drop_jurisdiction	0.703
drop_originality	0.634

Table 5: Bootstrap CIs for micro ProvScore; Wilcoxon on paired Δ .

Estimator	Mean	95% CI
BM25 micro-Prov	0.494	[0.487, 0.500]
PACER micro-Prov	0.690	[0.685, 0.695]
Δ (paired)	0.197	[0.189, 0.204]

and jurisdiction cues are partially redundant with authority and copy labels in the synthetic schema; the ablation nonetheless demonstrates that no single term is universally disposable.

7.4 Bootstrap and Wilcoxon

Table 5 reports bootstrap 95% confidence intervals on micro-averaged ProvScore. Paired gains are large ($\Delta=0.197$, CI [0.189, 0.204]) with Wilcoxon $z \approx 30$ ($p \approx 0$), consistent with McNemar on PSA@1.

7.5 Wikipedia revision slice

Table 6 summarizes the real revision track ($n=140$). PACER improves PSA@1, recall@ k , and nDCG relative to BM25, with McNemar $p=0.03125$. Absolute PSA remains low (0.057), reflecting long revision texts, noisy first-sentence gold, and programmatic labels that do not capture editor intent. We treat this track as an external stress test rather than a solved setting.

7.6 Pooled heterogeneous index

Table 7 combines synthetic and Wikipedia queries in one index. BM25 recall@ k falls to 0.435 and ProvScore to 0.272 when query styles and metadata conventions collide. PACER raises recall to 0.782 and ProvScore to 0.497, but heterogeneous deployment would still require domain routing or index partitioning in practice.

8 Discussion

What the benchmark shows. ProvBench-RAG separates recall of *some* supporting passage from selection of the *preferred* instance. Under lexical decoys, high recall masks systematic rank-1 provenance failure for BM25—a pattern that standard QA retrieval metrics under-report.

Limits of PACER. On synthetic data, PACER matches a strong authority-first heuristic, suggesting that much of the lift comes from re-ordering by authority once lexical scores are comparable within the pool. PACER’s value is interpretability and multi-signal extensibility (freshness, copy labels, jurisdiction) rather than a large margin over a single heuristic. Neural cross-encoders and learned fusion remain future work atop the same pools.

Table 6: Wikipedia revision freshness track ($n=140$).

Method	PSA@1	Recall@k	nDCG@k	Ans.Acc.	ProvScore
bm25	0.014	0.321	0.126	0.798	0.422
authority_first	0.014	0.321	0.126	0.798	0.422
recent_first	0.007	0.393	0.161	0.783	0.432
pacer	0.057	0.436	0.199	0.798	0.458
cluster_pacer	0.057	0.600	0.286	0.798	0.493

Table 7: Pooled synthetic + Wikipedia retrieval ($n=1640$).

Method	PSA@1	Recall@k	nDCG@k	Ans.Acc.	ProvScore
bm25	0.012	0.435	0.226	0.213	0.272
authority_first	0.236	0.435	0.338	0.251	0.338
recent_first	0.011	0.056	0.029	0.252	0.228
pacer	0.278	0.782	0.565	0.435	0.497
cluster_pacer	0.278	0.677	0.546	0.435	0.473

Threats to validity. Synthetic clusters, while controlled, share templated language; Wikipedia gold is programmatic; significance tests assume independent queries though clusters share templates; abstention is rule-based rather than crowd-adjudicated.

9 Ethics

Synthetic documents contain no personal data. Wikipedia crawling uses conservative request delays and public read-only APIs in line with Wikimedia guidance.

10 Reproducibility

`bash scripts/reproduce.sh` rebuilds synthetic metrics. `bash scripts/reproduce_full.sh` rebuilds Wikipedia, merged, and extended results. `python -m provbench_rag.scripts.export_paper_artifacts` refreshes `paper/generated_macros.tex` and tables.

11 Conclusion

Near-duplicate source confusion is a distinct retrieval failure mode for knowledge-augmented systems. ProvBench-RAG provides reproducible instance-level annotations and metrics to measure it, and PACER demonstrates that lightweight provenance fusion can recover preferred sources when lexical rankers fail—while also revealing substantial headroom on real revision corpora and heterogeneous indexes.

References

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